Covariance Matrix Estimation in High Dimension

A Statistical and Economic Comparison for Index Replication

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Abstract

The true covariance matrix of asset returns is never observable, and its natural estimator, the sample covariance, becomes unreliable when the number of assets p is of the same order as the number of observations n. This paper compares five estimators, the rolling sample covariance, EWMA (RiskMetrics), Ledoit–Wolf linear shrinkage (2004), analytical nonlinear shrinkage (ANLS, 2020) and quadratic-inverse shrinkage (QIS, 2022), along two complementary axes. The statistical axis measures the estimation error against a known true covariance in two simulated settings, one static and one dynamic. The economic axis measures the tracking error obtained when the estimators are used to replicate the MSCI EMU and MSCI USA indices, first on the historical path and then over 144,000 backtest scenarios that vary the exclusion rate, the estimation window, the data frequency and the rebalancing frequency. The ratio p/n emerges as the main source of estimation uncertainty. Shrinkage estimators, and QIS in particular, are much less sensitive to it than the sample covariance, and the resulting hierarchy QIS $>$ LW $>$ ANLS $>$ Rolling $>$ EWMA is stable across all configurations.

1. Introduction

The estimation of a covariance matrix inherently involves estimation uncertainty. In practice the true covariance of returns is never observable from real data, which makes any direct evaluation of the error made by an estimator difficult. Yet this matrix is at the core of portfolio construction: optimization, risk budgeting and index replication all depend on it, and in particular on its inverse.

The difficulty is a problem of dimension. A covariance matrix of p assets contains $p(p+1)/2$ parameters, all estimated from the same n observations. When p is of the same order as n, which is the normal situation for an equity index of several hundred stocks and a few years of data, the sample covariance becomes noisy, its eigenvalues are systematically distorted, and an optimizer exploits precisely the directions that are worst estimated.

The objective of this work is to compare different covariance estimation models in order to identify those that most faithfully approximate the unknown true matrix, and that lead to the best allocations in practice. Five estimators are compared: the rolling sample covariance, the exponentially weighted moving average of RiskMetrics, the linear shrinkage of Ledoit and Wolf (2004), the analytical nonlinear shrinkage of Ledoit and Wolf (2020) and the quadratic-inverse shrinkage of Ledoit and Wolf (2022).

The evaluation is conducted along two complementary axes. The *statistical* axis places the analysis in a controlled environment where the true covariance is specified by construction, so that the estimation error can be measured directly. The *economic* axis evaluates the estimators in a realistic portfolio management framework, index replication by tracking-error minimization, first on the historical path of two indices and then on a large set of scenarios that vary the operational parameters. The statistical evaluation analyzes the intrinsic quality of each estimator; the economic evaluation measures its concrete impact on a real portfolio.

The paper is organized as follows. Section 2 reviews the literature. Section 3 explains why the sample covariance fails in high dimension. Section 4 presents the estimators. Sections 5 to 7 report the statistical evaluation, the economic backtests and the empirical validation. Section 8 concludes.

2. Related literature

Three strands of literature frame this study.

Estimation error in optimization. Since Markowitz (1952), it is well known that optimized portfolios are highly sensitive to errors in their inputs; Michaud (1989) famously described optimizers as "error maximizers". Jagannathan and Ma (2003) showed that a long-only constraint already acts as an implicit regularization of the covariance matrix, which is relevant for the problem studied here. Work on tracking-error optimization (Roll, 1992; Jorion, 2003) generally takes the covariance matrix as given; this is precisely the assumption we relax.

Random matrix theory. Marčenko and Pastur (1967) showed that, when p/n does not vanish, the eigenvalues of the sample covariance matrix are systematically more dispersed than the true ones. Laloux et al. (1999) brought this result to finance and showed that most of the spectrum of empirical correlation matrices is indistinguishable from noise.

Shrinkage. Ledoit and Wolf (2004) proposed to correct this bias by pulling the sample matrix towards a scaled identity, with an intensity estimated from the data. Nonlinear shrinkage refines the idea by correcting each eigenvalue differently (Ledoit and Wolf, 2012); it was made analytical by Ledoit and Wolf (2020) and adapted to the precision matrix by Ledoit and Wolf (2022). Ledoit and Wolf (2017) showed its value for minimum-variance portfolios. Our study is closest to theirs but differs in three respects: the objective is tracking error, the investable universe is a strict subset of the benchmark, and the comparison is replicated over a large grid of operational configurations.

3. The estimation problem in high dimension

3.1. The true covariance and the natural estimator

We observe n daily returns of p assets. Let $r_t = (r_{1t}, \dots, r_{pt})^\top$ be the centered return vector at date t. The true covariance $\Sigma = \mathbb{E}[r_t r_t^\top]$ is a $p \times p$ symmetric positive definite matrix containing p variances and $p(p-1)/2$ covariances, i.e. $p(p+1)/2$ distinct parameters. It is unknown, we only observe the $n \times p$ matrix of returns X. The natural estimator is the sample covariance

$$S = \frac{1}{n} X^\top X, \qquad \hat{S}_{ij} = \frac{1}{n} \sum_{t=1}^{n} r_{it} r_{jt}, \tag{1}$$

each entry being the average of the observed cross-products.

3.2. Individual sampling uncertainty

Our sample is one realization among infinitely many: on n other days we would obtain a slightly different $\hat{S}_{ij}$. The law of large numbers guarantees $\hat{S}_{ij} \to \Sigma_{ij}$ as $n \to \infty$, but for finite n each estimate carries a sampling uncertainty. With i.i.d. observations, $\mathrm{Var}(\hat{S}_{ij}) = \frac{1}{n}\mathrm{Var}(r_{it} r_{jt})$, and for Gaussian returns Wick's theorem (Wick, 1950) gives $\mathbb{E}[(r_{it} r_{jt})^2] = \Sigma_{ii}\Sigma_{jj} + 2\Sigma_{ij}^2$, hence

$$\mathrm{Var}(\hat{S}_{ij}) = \frac{\Sigma_{ii}\Sigma_{jj} + \Sigma_{ij}^2}{n}. \tag{2}$$

For a diagonal term this gives $2\Sigma_{ii}^2/n$, and for uncorrelated assets $\Sigma_{ii}\Sigma_{jj}/n$. The uncertainty on each entry decreases with n.

3.3. The total error: quadratic accumulation in p

All entries are estimated simultaneously, each with its own uncertainty, and the total error is the sum of these uncertainties, $\mathcal{E}_{\text{tot}} = \mathbb{E}\|S - \Sigma\|_F^2 = \sum_{i,j} \text{Var}(\hat{S}_{ij})$. With a common variance σ^2 and an average correlation ρ,

$$\mathcal{E}_{\text{tot}} = \underbrace{\frac{2p\sigma^4}{n}}_{\text{diagonal}} + \underbrace{p(p-1)\,\frac{\sigma^4(1+\rho^2)}{n}}_{\text{off-diagonal}} \tag{3}$$

and, since the p^2 off-diagonal terms dominate for large p,

$$\mathcal{E}_{\text{tot}} \approx \frac{p^2\sigma^4(1+\rho^2)}{n} = O\left(\frac{p^2}{n}\right). \tag{4}$$

The total error is quadratic in p and linear in $1/n$: if the number of assets doubles, the error quadruples.

3.4. Eigenvalues and why they matter

In finance, S is used through global matrix operations (Markowitz optimization, minimum variance, tracking error), all of which go through its spectral decomposition. An eigenvector u_k and its eigenvalue λ_k satisfy $\Sigma u_k = \lambda_k u_k$, and the eigenvalue has a direct financial interpretation: it is the variance of the portfolio u_k, since for $\|u_k\| = 1$

$$\text{Var}(u_k^\top r) = u_k^\top \Sigma u_k = \lambda_k. \tag{5}$$

The spectral theorem decomposes $\Sigma = \sum_k \lambda_k u_k u_k^\top$ into p independent "pure" risks, and its inverse, at the core of Markowitz, is

$$\Sigma^{-1} = \sum_{k=1}^{p} \frac{1}{\lambda_k}\, u_k u_k^\top. \tag{6}$$

Inversion replaces each λ_k by $1/\lambda_k$. If a small eigenvalue is underestimated, $1/\lambda_k$ becomes very large: the apparently low-risk direction receives an enormous and erroneous weight in the optimization.

3.5. How entry-level uncertainty propagates to the spectrum

Eigenvalues are nonlinear functions of all entries at once: computing them amounts to solving $\det(S - \lambda I) = 0$, a polynomial of degree p that mixes all $p(p+1)/2$ entries. Let $E = S - \Sigma$ be the error matrix. Weyl's theorem (Weyl, 1912) bounds the displacement of every eigenvalue:

$$|\ell_k(S) - \lambda_k(\Sigma)| \le \|E\|_{\text{op}}, \qquad \|E\|_{\text{op}} = \max_{\|u\|=1} |u^\top E u|, \tag{7}$$

where ℓ_k denotes the eigenvalues of S and the operator norm is the largest absolute eigenvalue of E. Each entry of E has a standard deviation of order $\sigma^2/\sqrt{n}$, which tends to zero. But the maximum is taken over all unit vectors of a p-dimensional space, which finds the direction in which the p^2 errors accumulate most:

$$\|E\|_{\text{op}} \approx \frac{\sigma^2}{\sqrt{n}}\sqrt{p} = \sigma^2\sqrt{c}, \qquad c = \frac{p}{n}, \tag{8}$$

which does not tend to zero if $c > 0$. **The more assets there are, the larger the unit sphere, and the more likely it is to find a direction that concentrates the noise.**

3.6. The Marčenko–Pastur law

Marčenko and Pastur (1967) quantified this bias exactly in the reference case $\Sigma = I$: all true eigenvalues equal one, so any dispersion of the eigenvalues of S is pure sampling noise. When $n, p \to \infty$ with $p/n \to c$, their distribution converges to

$$f_{\mathrm{MP}}(\lambda; c) = \frac{1}{2\pi c\lambda}\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}, \qquad \lambda \in [\lambda_-, \lambda_+], \quad \lambda_\pm = (1 \pm \sqrt{c})^2, \tag{9}$$

with an atom of mass $1 - 1/c$ at zero when $c > 1$ (the null eigenvalues of a singular matrix). The extreme eigenvalues converge to the edges of the support (Yin et al., 1988; Bai and Yin, 1993):

$$\ell_{\max}(S) \to (1 + \sqrt{c})^2 \qquad \text{(systematic overestimation)}, \tag{10}$$
$$\ell_{\min}(S) \to (1 - \sqrt{c})^2 \qquad \text{(systematic underestimation)}, \tag{11}$$

and all intermediate eigenvalues are distorted according to f_{MP}. The small eigenvalue, underestimated, produces an enormous $1/\ell_{\min}$ in S^{-1}, which is exactly the failure mode described above.

4. Covariance estimators

Table 1 lists the five estimators. The first two are direct estimators, the last three correct the spectrum of S.

Table 1. Estimators compared

Acronym	Estimator	Reference
Rolling	Sample covariance over a rolling window	—
EWMA	Exponentially weighted moving average, $\lambda = 0.94$	J.P. Morgan/Reuters (1996)
LW	Linear shrinkage towards a spherical target	Ledoit and Wolf (2004)
ANLS	Analytical nonlinear shrinkage	Ledoit and Wolf (2020)
QIS	Quadratic-inverse shrinkage	Ledoit and Wolf (2022)

4.1. Rolling sample covariance and EWMA

The *rolling* estimator is the sample covariance over a window of w days. It is non-parametric, requires no structural assumption and serves as the reference. The *EWMA* estimator computes the covariance by the RiskMetrics recursion $\hat{\Sigma}_t = \lambda\hat{\Sigma}_{t-1} + (1-\lambda)r_t r_t^\top$: all past observations contribute with exponentially decaying weights, which makes it responsive to regime changes without abrupt cutoffs. The decay is set to the standard RiskMetrics value $\lambda = 0.94$ for daily returns.

4.2. Linear shrinkage: Ledoit and Wolf (2004)

The idea of Ledoit and Wolf is to summarize the global bias of the spectrum by a single scalar and to correct it in one operation, by replacing S with a convex combination of S and a simple, well-conditioned target T:

$$\hat{\Sigma} = \rho T + (1 - \rho)S, \qquad \rho \in [0, 1]. \tag{12}$$

The target is the spherical matrix $T = \bar{s}I_p$, with $\bar{s} = \mathrm{tr}(S)/p$ the average variance. It is chosen for three reasons: it is perfectly conditioned (condition number one), it depends on a single estimated parameter, and it is the best approximation of Σ of the form αI in the least-squares

sense. It assumes that all assets have the same variance and are uncorrelated, a strong and probably false assumption, but one that introduces almost no estimation noise. Shrinkage trades a known bias for a large reduction in noise, and ρ doses this trade-off.

The optimal intensity minimizes $\mathbb{E}\|\rho T + (1-\rho)S - \Sigma\|_F^2$, which gives

$$\rho^* = -\frac{\mathbb{E}\langle T - S,\, S - \Sigma\rangle_F}{\mathbb{E}\|T - S\|_F^2}. \tag{13}$$

The authors show that it can be estimated from the data with two quantities. The first, $d^2 = \|S - T\|_F^2 = \sum_i (S_{ii} - \bar{s})^2 + \sum_{i \neq j} S_{ij}^2$, is the total dispersion of S around the target, signal and noise combined. The second, b^2, is the sampling variance of S, i.e. the part of d^2 due to noise, it measures how much the rank-one contributions $x_t x_t^\top$ of each observation fluctuate around their mean S:

$$\bar{b}^2 = \frac{1}{n^2}\sum_{t=1}^{n} \|x_t x_t^\top - S\|_F^2, \qquad b^2 = \min(\bar{b}^2, d^2). \tag{14}$$

The central result is

$$\boxed{\rho^* = \frac{b^2}{d^2}}, \qquad \hat{\Sigma}_{\mathrm{LW}} = \rho^* \bar{s} I_p + (1 - \rho^*) S, \tag{15}$$

the share of sampling noise in the total dispersion of S, the truncation $b^2 \leq d^2$ ensures $\rho^* \in [0, 1]$. If $b^2 \approx d^2$, almost all the dispersion is noise and shrinkage is strong, if $b^2 \ll d^2$, it is mostly signal and shrinkage is weak. At the spectral level, each eigenvalue becomes $\rho^* \bar{s} + (1 - \rho^*)\ell_k$: large (overestimated) eigenvalues are reduced, small (underestimated) ones increased.

The estimator is positive definite even when S is singular, costs $O(np^2)$ like S itself, is asymptotically equivalent to the oracle that would know Σ, and vanishes ($\rho^* \to 0$) in the classical regime $n \to \infty$ with p fixed. Its limitation is that ρ^* is a single scalar: all eigenvalues are pulled towards $\bar{s}$ with the same strength, whereas the bias depends on the position in the spectrum, large eigenvalues being inflated by $(1 + \sqrt{c})^2$ and small ones deflated by $(1 - \sqrt{c})^2$. A uniform correction cannot fix both biases optimally.

4.3. Analytical nonlinear shrinkage: Ledoit and Wolf (2020)

ANLS replaces the uniform correction by an individual correction of each eigenvalue. It keeps the eigenvectors of S, trusted for the directions, and replaces only the eigenvalues:

$$\hat{\Sigma}_{\mathrm{ANLS}} = U \mathrm{diag}(d_1, \ldots, d_p) U^\top, \tag{16}$$

where the d_k minimize $\mathbb{E}\|\sum_k d_k u_k u_k^\top - \Sigma\|_F^2$. Using the Stieltjes transform of the spectrum, whose limit near the real axis gives both the density f of the eigenvalues and its Hilbert transform Hf, the authors obtain each d_k in closed form:

$$d_k^* = \frac{\ell_k}{\big(\pi c\, \ell_k f(\ell_k)\big)^2 + \big(1 - c - \pi c\, \ell_k\, Hf(\ell_k)\big)^2}. \tag{17}$$

The correction depends on two quantities for each eigenvalue: the local density $f(\ell_k)$, which gives the amplitude, where eigenvalues are crowded, noise is strong and more correction is needed, and the Hilbert transform $Hf(\ell_k)$, which captures the local asymmetry of the spectrum, whether more eigenvalues lie to the left or to the right of ℓ_k and gives the direction of the correction.

Both are estimated by kernel density estimation on the p sample eigenvalues, with the Epanechnikov kernel (Epanechnikov, 1969), optimal among compactly supported kernels, and a bandwidth of order $h = n^{-1/3}$:

$$K(u) = \frac{3}{4\sqrt{5}}\Big(1 - \frac{u^2}{5}\Big)^+, \qquad HK(u) = -\frac{3u}{10\pi} + \frac{3}{4\sqrt{5}\,\pi}\Big(1 - \frac{u^2}{5}\Big)\ln\left|\frac{\sqrt{5}-u}{\sqrt{5}+u}\right|. \tag{18}$$

The key property is that the Hilbert transform of the kernel is known in closed form, so Hf is obtained from the same sum as f with HK in place of K. When $p > n$, the $p - n$ null eigenvalues of S receive a common value computed from the Hilbert transform of the density at zero.

ANLS is asymptotically optimal among estimators that keep the sample eigenvectors, adapts the correction to each eigenvalue and is positive definite. It has two limitations: it minimizes the error on Σ, whereas finance uses Σ^{-1}, where a small error on an eigenvalue close to zero becomes a large error, and its compact kernel introduces a boundary bias, extreme eigenvalues being poorly estimated because the kernel spills outside the true support of the spectrum.

4.4. Quadratic-inverse shrinkage: Ledoit and Wolf (2022)

QIS changes the optimization criterion. Since an additive error ε on an eigenvalue d_k becomes an error of ε/d_k^2 on its inverse, inversion amplifies errors quadratically on small eigenvalues. QIS therefore minimizes directly the error on the precision matrix,

$$\min_{d_1,\dots,d_p} \mathbb{E}\Big\|\Big(\sum_k d_k u_k u_k^\top\Big)^{-1} - \Sigma^{-1}\Big\|_F^2, \tag{19}$$

and works on the inverse eigenvalues $\mu_k = 1/\ell_k$, in which the low-variance directions (the most poorly estimated) become the large values and naturally receive more attention.

It uses the Cauchy kernel, whose infinite support suits the very dispersed precision spectrum and whose Hilbert transform is exact. For each μ_i, QIS computes the smoothed Stein shrinker and its conjugate over the $m = \min(p, n)$ nonzero precisions,

$$\hat\theta_i = \frac{1}{m}\sum_{j=1}^{m}\frac{\mu_j(\mu_j - \mu_i)}{(\mu_j-\mu_i)^2 + h^2\mu_j^2}, \qquad \widehat{H\theta}_i = \frac{1}{m}\sum_{j=1}^{m}\frac{h\mu_j^2}{(\mu_j-\mu_i)^2 + h^2\mu_j^2}, \tag{20}$$

which measure respectively the attraction of μ_i towards its neighbors (its relative position) and the local density of precisions around it, and their squared amplitude $\hat A_i^2 = \hat\theta_i^2 + \widehat{H\theta}_i^2$, the overall intensity of noise around μ_i. The bandwidth $h = \min(c^2, 1/c^2)^{0.35}/p^{0.35}$ adapts to the dimensional regime, unlike the $h = n^{-1/3}$ of ANLS.

The corrected eigenvalues are given by a formula quadratic in c:

$$\frac{1}{d_k^*} = \underbrace{(1-c)^2\mu_k}_{c\to 0} + \underbrace{2c(1-c)\,\mu_k\hat\theta_k}_{c\approx 0.5} + \underbrace{c^2\mu_k\hat A_k^2}_{c\to 1}, \qquad \text{i.e.} \qquad d_k^* = \frac{\ell_k}{(1-c)^2 + 2c(1-c)\hat\theta_k + c^2\hat A_k^2}. \tag{21}$$

The three weights sum to one: the raw precision dominates in the classical regime, the Stein shrinker at intermediate c, and the amplitude correction as $c \to 1$. When $c = 0$, $d_k^* = \ell_k$ and there is no correction. When $p > n$, the null directions receive the common value $1/[(c-1)\bar\mu]$, with $\bar\mu$ the mean of the nonzero precisions. Finally, the eigenvalues are rescaled so that $\mathrm{tr}(\hat\Sigma_{\mathrm{QIS}}) = \mathrm{tr}(S)$: since the formula optimizes the error on Σ^{-1}, nothing guarantees that total variance is preserved, and the rescaling prevents an uncontrolled change in the overall volatility of portfolios. QIS is thus optimized for inversion, keeps the eigenvectors of S, is positive definite, and avoids the boundary bias of the Epanechnikov kernel.

5. Statistical evaluation

5.1. Protocol

Synthetic returns are generated from a perfectly known true covariance, each estimator produces an estimate, and the gap with the truth is measured. The true covariance is entirely synthetic. Only the average daily volatility is fixed at a realistic 1%. Each experiment is repeated over many independent scenarios (random seeds), and we report the mean losses. Parameters are given in Section A.

DGP 1: Static Oracle. A single true covariance with factor structure, $\Sigma = BB^\top + \mathrm{diag}(\sigma^2_{\mathrm{idio}})$, is drawn per scenario, with $B \in \mathbb{R}^{p\times 10}$ Gaussian loadings and log-normal idiosyncratic variances. We then simulate T i.i.d. returns $r_t = \mathrm{chol}(\Sigma)z_t$, with z_t Gaussian or Student, and each estimator produces one matrix $\hat{\Sigma}$. Two regimes are tested: $p/T = 0.2$ (easy, S well-conditioned) and $p/T = 1$ (hard, S at the boundary of singularity).

DGP 2: Factor Shock. The true covariance now evolves at each date. Loadings and log-idiosyncratic variances follow stationary AR(1) processes,

$$B_t = \rho_B B_{t-1} + \sqrt{1-\rho_B^2}\,\sigma_B \varepsilon_t^B, \qquad \ell d_t = \rho_d\, \ell d_{t-1} + \sqrt{1-\rho_d^2}\,\sigma_d \varepsilon_t^d, \tag{22}$$

with $\Sigma_t = B_t B_t^\top + \mathrm{diag}(e^{\ell d_t})$ and $r_t = \mathrm{chol}(\Sigma_t)z_t$. The calibration ($\rho_B = 0.95$, $\sigma_B = 0.05$, $\rho_d = 0.97$, $\sigma_d = 0.10$, $K = 10$) produces a covariance that is persistent but not constant, as in real markets. Each estimator produces a sequence $\{\hat{\Sigma}_t\}$ on a rolling window $w \in \{252, 504\}$, and losses are averaged over the evaluation dates, for $N_{\mathrm{sim}} \in \{100, 500\}$ assets.

Loss metrics. Three losses are computed for each pair $(\hat{\Sigma}, \Sigma)$. The *Frobenius norm* $\|\hat{\Sigma} - \Sigma\|_F$ measures the total error over all entries. The *spectral norm*, the largest absolute eigenvalue of $\hat{\Sigma} - \Sigma$, measures the variance error on the most unfavorable portfolio and is sensitive to errors on the dominant directions. The *Stein loss*,

$$L_{\mathrm{Stein}}(\hat{\Sigma}, \Sigma) = \mathrm{tr}(\hat{\Sigma}\Sigma^{-1}) - \log\det(\hat{\Sigma}\Sigma^{-1}) - p = \sum_i \Big(\frac{\hat{\lambda}_i}{\lambda_i} - \log\frac{\hat{\lambda}_i}{\lambda_i} - 1\Big), \tag{23}$$

corresponds up to a constant to the Kullback–Leibler divergence between two Gaussian distributions. It penalizes relative errors on each eigenvalue and is particularly sensitive to small eigenvalues, which drive optimized portfolios.

5.2. Results: Static Oracle

Under Gaussian innovations the theoretical hierarchy is respected almost perfectly (Table 2): QIS dominates on all metrics in both regimes, followed by LW. The gap is largest on the Stein loss, 13.63 against 26.51 at $p/T = 0.2$, and 42.22 against 200.98 at $p/T = 1$ and grows with p/T, which illustrates the superiority of QIS in underdetermined regimes. The absolute levels are low: a Frobenius norm of 0.0006–0.0007 means that LW and QIS estimate the covariances between asset pairs very precisely, which is directly relevant for risk budgeting.

Student innovations raise the Frobenius losses of LW and QIS by a factor of 6 to 7, but absolute values remain low. Their main effect is on the Stein loss: the heavy tails produce outliers that distort the small eigenvalues, and LW deteriorates sharply (1250.47 at $p/T = 1$, against 200.98 under Gaussian innovations), while QIS resists much better (131.80) and keeps its lead on the Stein loss in both regimes. Rolling is systematically dominated by the shrinkage

Table 2. Static Oracle: mean losses by innovation type (10,000 scenarios each)

	$p/T = 0.2$			$p/T = 1.0$		
Model	Frobenius	Spectral	Stein	Frobenius	Spectral	Stein
Gaussian innovations						
ANLS	0.0037	0.0018	44.74	0.0035	0.0023	$\gg 10^9$
EWMA	0.0036	0.0023	842.75	0.0035	0.0023	261.48
LW	0.0007	0.0004	26.51	0.0014	0.0008	200.98
QIS	**0.0006**	**0.0003**	**13.63**	**0.0012**	**0.0007**	**42.22**
Rolling	0.0013	0.0008	119.49	0.0015	0.0008	203.99
Student-t_5 innovations						
ANLS	0.0105	0.0053	169.72	0.0087	0.0071	$\gg 10^9$
EWMA	0.0093	0.0075	932.92	0.0087	0.0071	339.88
LW	**0.0042**	**0.0024**	290.46	**0.0043**	**0.0030**	1250.47
QIS	0.0044	**0.0024**	**69.94**	0.0051	0.0033	**131.80**
Rolling	0.0054	0.0036	201.47	0.0057	0.0039	294.78

Note. True variances are of order $(0.01)^2 = 10^{-4}$. Bold: best in column.

estimators. EWMA has the highest Stein losses: its weighting, designed to track a moving covariance, is poorly suited to a fixed one. Finally, at $p/T = 1$ the Stein loss of ANLS becomes numerically unstable: at the boundary of singularity the smallest eigenvalues tend to zero and the terms of the Stein loss explode for estimators that do not regularize the spectrum enough.

5.3. Results: Factor Shock

The dynamic framework modifies the hierarchy (Table 3). LW now dominates on all metrics and both windows: under Gaussian innovations and $N_{\text{sim}} = 100$, its Stein loss is 0.72 against 1.10 for QIS. A possible explanation is that, with strong persistence ($\rho_B = 0.95$), the true covariance evolves slowly along a linear process, which favors linear shrinkage, whereas the more aggressive spectral correction of QIS slightly over-regularizes. LW and QIS nonetheless jointly dominate all other models.

The increase in dimension from $N_{\text{sim}} = 100$ to 500 strongly degrades the non-shrinkage models: the Stein loss of Rolling 252 rises from 24.22 to 3343.98, and that of EWMA from 200 to 4341, whereas LW only moves from 0.72 to 5.13. This confirms the direct exposure of these estimators to the p/n ratio. ANLS with a 504-day window produces aberrant values at $N_{\text{sim}} = 500$, revealing a numerical instability of this estimator in this high-dimensional dynamic configuration. Student innovations degrade all models, LW remaining the most robust on Frobenius and spectral norms.

5.4. Summary of the statistical evaluation

(i) **QIS dominates in the static i.i.d. framework**, especially on the Stein loss, with an advantage that grows with p/T and resists heavy tails.

(ii) **LW takes the advantage in the dynamic framework**, where the covariance is persistent, LW and QIS jointly dominate all other models.

(iii) **Rolling and EWMA are structurally exposed to the p/n ratio**: their Stein loss explodes with dimension.

(iv) **EWMA is the least robust model**, and **ANLS with a long window is numerically unstable** at large dynamic dimension.

Table 3. Factor Shock: mean losses by innovation type and dimension (1,000 scenarios each)

	Gaussian				Student-t_5			
	$N_{\text{sim}} = 100$		$N_{\text{sim}} = 500$		$N_{\text{sim}} = 100$		$N_{\text{sim}} = 500$	
Model	Frob.	Stein	Frob.	Stein	Frob.	Stein	Frob.	Stein
ANLS 252	0.0024	99.54	0.0029	216.14	0.0057	249.47	0.0083	512.79
ANLS 504	0.0009	23.55	45.77	7.7×10^6	0.0024	87.17	65.67	1.0×10^7
EWMA 252	0.0017	200.46	0.0086	4341.29	0.0041	242.10	0.0201	4609.07
EWMA 504	0.0017	199.25	0.0086	3749.99	0.0041	240.14	0.0203	3981.94
LW 252	0.0001	0.72	0.0004	5.13	0.0007	17.41	0.0015	91.64
LW 504	0.0001	**0.72**	0.0004	**5.11**	0.0007	**17.39**	0.0015	**91.19**
QIS 252	0.0001	1.10	0.0005	6.64	0.0012	26.30	0.0051	193.12
QIS 504	0.0001	0.91	0.0005	12.98	0.0010	23.29	0.0039	174.26
Rolling 252	0.0006	24.22	0.0031	3343.98	0.0018	56.47	0.0082	3631.81
Rolling 504	0.0005	11.49	0.0022	492.28	0.0013	37.61	0.0060	726.48

Note. Losses averaged over the evaluation dates of each path. Bold: lowest Stein loss in column. Spectral norms follow the same ranking.

6. Economic evaluation

6.1. Data and backtest framework

The study covers the MSCI EMU (Bloomberg ticker MSDEEMUN, about 240 stocks) and the MSCI USA (M1US, about 600 stocks). Daily prices are sourced from Bloomberg from 2 January 2017 to 31 December 2025, and backtests run over 2020–2025. At each rebalancing date only the securities actually in the index are used, which excludes survivorship bias. Benchmark weights are the quarterly capitalization weights, drifted with the returns of the constituents between two rebalancings and renormalized. Dividends and corporate actions are not adjusted, since all models are evaluated on the same price basis, this does not affect the comparison.

At each rebalancing date τ, the portfolio minimizes the ex-ante tracking error against the benchmark b_τ:

$$w_\tau^* = \arg\min_{w \in \mathcal{C}} (w - b_\tau)^\top \hat{\Sigma}_\tau (w - b_\tau), \tag{24}$$

where $\mathcal{C}$ imposes full investment ($\mathbf{1}^\top w = 1$), no short positions ($w \geq 0$) and restriction to the investable universe. $\hat{\Sigma}_\tau$ is estimated on data up to $\tau - 1$, so the procedure is causal. The problem is solved by Clarabel (Goulart and Chen, 2024), which guarantees global optimality. Rebalancing is quarterly (index rebalancing date plus one day), transaction costs are zero, and windows of 252, 504 and 756 days are tested.

To reflect practical constraints, part of the index is excluded from the investable universe (responsible-investment or internal policies), while the target remains the full index. This asymmetry creates a structural tracking error that the optimizer must minimize using the covariance information. Two universes are therefore studied for each index: the full universe, and a restricted universe excluding on average 3.6% of the capitalization (MSCI EMU) and 8.8% of the constituents (MSCI USA).

Metrics. The main metric is the annualized ex-post tracking error, $\text{TE}_{\text{ex-post}} = \sqrt{252}\,\sigma(r_t^p - r_t^b)$. We also report the cumulative active performance.

Table 4. Annualized ex-post tracking error (%), 2020–2025

	MSCI EMU (≈ 240 stocks)						MSCI USA (≈ 600 stocks)					
	Without exclusions			With exclusions			Without exclusions			With exclusions		
Window	252	504	756	252	504	756	252	504	756	252	504	756
Rolling	0.10	0.10	0.10	0.81	0.58	0.54	0.34	0.19	0.19	0.87	0.54	0.49
EWMA	0.22	0.22	0.20	0.95	0.95	0.92	0.39	0.39	0.35	0.77	0.78	0.71
LW	0.09	0.09	0.10	0.81	0.53	0.52	0.17	0.17	0.18	0.53	0.49	0.44
ANLS	0.10	0.10	0.10	0.81	0.58	0.54	0.17	0.17	0.18	0.58	0.63	0.57
QIS	0.09	0.09	0.10	0.58	0.54	0.52	0.17	0.17	0.18	0.53	0.50	0.49
Cumulative active performance (%)												
Rolling	0.14	0.14	0.20	−0.47	−0.27	−0.36	0.55	0.37	0.28	1.95	2.36	1.66
EWMA	0.18	0.17	−0.05	−0.88	−0.88	−1.50	1.07	1.09	0.13	2.13	2.04	1.31
LW	0.16	0.15	0.22	−0.43	0.40	0.22	0.49	0.53	0.28	1.88	3.23	1.46
ANLS	0.15	0.14	0.20	−0.43	−0.27	−0.36	0.49	0.54	0.27	1.85	2.86	1.70
QIS	0.17	0.18	0.22	0.29	0.50	0.22	0.47	0.50	0.28	1.98	2.11	1.66

Note. Quarterly rebalancing, daily data, zero transaction costs.

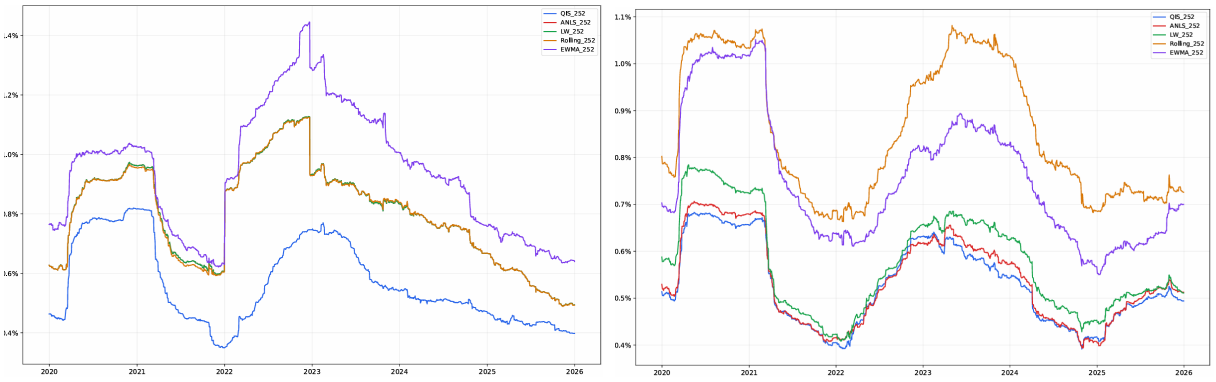
6.2. Results

MSCI EMU without exclusions. Tracking errors are low and very close across models, between 0.09% and 0.10% (Table 4). LW and QIS are slightly ahead at 0.09% on the short windows, and all models converge at 756 days. EWMA is the exception, at about 0.22%. Without exclusions, near-direct replication is attainable: the tracking error comes mainly from estimation noise and optimization constraints, and the role of the estimator remains limited.

MSCI EMU with exclusions. Exclusions profoundly change the level and the discrimination between models. On the 252-day window, where p/n is close to 1, Rolling, ANLS and LW all reach 0.81% while QIS stands out at 0.58%, demonstrating its ability to stabilize the estimation in a very noisy setting (Figure 1a). Lengthening the window lowers p/n and improves all models except EWMA: Rolling moves from 0.81% to 0.54%, and at 756 days LW and QIS reach 0.52% against 0.54% for Rolling and ANLS. EWMA remains around 0.92–0.95%. Here the gaps are structural: the impossibility of replicating the index directly places the estimator at the center of the problem.

MSCI USA without exclusions. With about 600 stocks, p/n is close to 2.4 on the 252-day window, which discriminates the models more clearly. Rolling reaches 0.34%, twice the 0.17% of the shrinkage models, a gap of 17 basis points that illustrates the stability gain of shrinkage in high dimension. Lengthening the window benefits Rolling most (0.19% at 504 and 756 days), but it never equals the shrinkage models. EWMA stays at 0.35–0.39%.

MSCI USA with exclusions. With exclusions, the gaps become persistent and economically significant. At 252 days, LW and QIS reach 0.53%, ANLS 0.58%, EWMA 0.77% and Rolling 0.87% (Figure 1b). With longer windows, LW stands out with 0.49% at 504 days and 0.44% at 756 days, the best level of all models, while ANLS does not benefit from lengthening (0.57–0.63%) and approaches Rolling. EWMA remains between 0.71% and 0.78%.



(a) MSCI EMU with exclusions **(b)** MSCI USA with exclusions

Figure 1. Ex-post tracking error (rolling 252 days, annualized), 252-day estimation window.

6.3. Summary of the economic backtest

The conclusions are consistent across indices and configurations. First, the p/n ratio plays a central role: short windows produce higher and more dispersed tracking errors, and lengthening the window brings the models closer, particularly benefiting Rolling. Second, shrinkage models keep a clear and persistent advantage, even in the most unfavorable settings. Third, the hierarchy is stable:

$$\{\text{QIS, LW}\} > \text{ANLS} > \text{Rolling} > \text{EWMA},$$

QIS leading on short windows and in the noisiest regimes, LW on long windows. This analysis, however, relies on a single market trajectory, which motivates the validation of the next section.

7. Empirical validation

7.1. Protocol

A single realization is not sufficient to establish robust conclusions: a favorable result may reflect a better fit to the particular period rather than an intrinsic superiority. We therefore keep the true return data but systematically vary the parameters of the experiment (Table 5). For each exclusion rate, 100 investable universes are generated by random exclusion, and each is backtested for every window, data frequency and rebalancing frequency. This yields 18,000 scenarios per model and index, 72,000 per index and 144,000 in total. EWMA is excluded from this phase to reduce computation time, its unsuitability having been established in the previous sections.

Table 5. Scenario grid of the empirical validation

Dimension	Values	Count
Exclusion rate f	5, 10, 20, 30, 40, 50%	6
Random universes per rate		100
Estimation window	126, 252, 378, 504, 756 days	5
Data frequency	daily, weekly	2
Rebalancing frequency	weekly, monthly, quarterly	3
Scenarios per model and index	$600 \times 5 \times 2 \times 3$	18,000

Table 6. Ex-post tracking error over all scenarios (%)

	MSCI EMU				MSCI USA			
Model	Mean	Std	Min	Max	Mean	Std	Min	Max
ANLS	1.45	0.71	0.17	3.51	1.60	0.89	0.18	4.83
LW	1.38	0.71	0.16	3.57	1.57	0.88	0.18	4.28
QIS	**1.35**	0.70	0.16	3.57	**1.53**	0.87	0.18	4.50
Rolling	1.81	0.81	0.19	5.42	2.03	0.92	0.23	4.98

Note. 18,000 scenarios per model and index.

7.2. Aggregated results

The hierarchy QIS > LW > ANLS > Rolling holds on both universes (Table 6). QIS achieves the best mean (1.35% and 1.53%), the gaps between the three shrinkage models are limited (5 to 10 basis points), and Rolling is 46 and 50 basis points above QIS, with the highest dispersion and maximum (5.42% on the MSCI EMU), confirming its sensitivity to unfavorable configurations.

7.3. Effect of the operational parameters

Data frequency. Switching from daily to weekly data increases the mean tracking error by 21 basis points on the MSCI EMU and 25 on the MSCI USA (Table 7, Panel A), because fewer observations over the same calendar window mechanically raise p/n. The effect is asymmetric: the shrinkage models lose only 3 to 14 basis points, their regularization partially compensating for the loss of observations, whereas Rolling loses about 60 basis points on both universes. On liquid large-cap universes, microstructure frictions are limited, so the benefit of weekly smoothing is small compared with the cost of fewer observations: daily data are preferable.

Estimation window. Lengthening the window improves the mean tracking error monotonically, by 14 basis points on the MSCI EMU and 27 on the MSCI USA between 126 and 756 days (Panel B). The effect is very different across models. Shrinkage models show small differences between their best and worst windows (7 to 17 basis points), their regularization compensating the p/n bias even on short windows. Rolling is far more sensitive: it gains 37 basis points on the MSCI EMU and 65 on the MSCI USA, without ever reaching the level of the shrinkage models.

Exclusion rate. The exclusion rate is the main driver of the absolute level of tracking error (Panel C): the mean moves from 0.66% at $f = 5\%$ to 2.43% at $f = 50\%$ on the MSCI EMU, and from 0.68% to 2.71% on the MSCI USA, a factor close to four. The model hierarchy is preserved at every exclusion level, QIS dominating throughout, with a gap to Rolling of about 44–45 basis points on the MSCI EMU and 43–58 basis points on the MSCI USA.

Rebalancing frequency. Rebalancing frequency is a secondary parameter (Panel D): the three frequencies produce nearly identical tracking errors, the maximum gap being 1 basis point on the MSCI EMU and 5 on the MSCI USA, and the hierarchy is unchanged. Turnover, however, more than doubles from quarterly (about 1.5 per year) to weekly rebalancing (3.3 on the MSCI EMU, 3.9 on the MSCI USA). Quarterly rebalancing therefore offers the best tracking-error/turnover trade-off; with realistic transaction costs, the marginal gain of weekly rebalancing would likely be eliminated.

Table 7. Mean ex-post tracking error (%) by operational parameter

	MSCI EMU				MSCI USA			
	QIS	LW	ANLS	Rolling	QIS	LW	ANLS	Rolling
Panel A. Data frequency								
Daily	1.30	1.33	1.43	1.52	1.46	1.51	1.54	1.72
Weekly	1.40	1.43	1.46	2.11	1.60	1.63	1.66	2.35
Panel B. Estimation window (days)								
126	1.39	1.43	1.47	1.94	1.64	1.63	1.69	2.41
252	1.36	1.40	1.49	1.96	1.55	1.58	1.61	2.13
378	1.35	1.38	1.42	1.87	1.51	1.56	1.59	1.97
504	1.34	1.36	1.41	1.72	1.49	1.54	1.58	1.88
756	1.32	1.34	1.44	1.59	1.47	1.53	1.53	1.76
Panel C. Exclusion rate								
5%	0.53	0.55	0.61	0.97	0.52	0.54	0.56	1.10
10%	0.74	0.76	0.83	1.20	0.88	0.92	0.94	1.42
20%	1.20	1.23	1.30	1.68	1.40	1.44	1.47	1.90
30%	1.52	1.55	1.62	1.99	1.75	1.80	1.82	2.24
40%	1.84	1.88	1.94	2.31	2.06	2.11	2.15	2.53
50%	2.29	2.33	2.38	2.74	2.57	2.60	2.66	3.00
Panel D. Rebalancing frequency								
Weekly	1.34	1.37	1.43	1.80	1.51	1.55	1.58	2.01
Monthly	1.36	1.39	1.45	1.82	1.55	1.59	1.62	2.07
Quarterly	1.35	1.39	1.45	1.82	1.53	1.57	1.60	2.01

7.4. Robustness and stress behavior

All models see their tracking error slightly decrease in bear markets (ratio below one, Table 8): during downturns correlations increase and converge, which makes the covariance easier to estimate. The ratios are very close across models, so the differentiation lies in the absolute level, where QIS $<$ LW $<$ ANLS $<$ Rolling. QIS has the most stable annual tracking error on both universes, while Rolling is the least stable and has the deepest mean active drawdowns (-5.65% and -6.11%).

Finally, the dispersion of tracking error across scenarios is comparable for all models (coefficients of variation between 0.45 and 0.57). This variability mainly reflects the differences between the generated investable universes rather than any instability of the models: the choice of excluded assets is at least as important as the choice of estimator in explaining the tracking error of a given scenario.

8. Conclusion

This study compared five covariance estimators along a statistical axis, where the true covariance is known, and an economic axis, through index replication backtests on real data extended by 144,000 scenarios. The results organize around a single principle: **the p/n ratio is the main source of estimation uncertainty**. When it is high, non-regularized estimators degrade strongly, while shrinkage estimators keep stable performance.

The hierarchy is consistent across all frameworks:

$$\text{QIS} \;>\; \text{LW} \;>\; \text{ANLS} \;>\; \text{Rolling} \;>\; \text{EWMA}.$$

Table 8. Risk profile by model

	MSCI EMU				MSCI USA			
Model	Bear TE	Ratio	Stability	Max DD	Bear TE	Ratio	Stability	Max DD
ANLS	1.40	0.964	0.29	−4.88	1.58	0.985	0.31	−5.28
LW	1.34	0.971	0.28	−4.72	1.54	0.982	0.30	−4.97
QIS	1.31	0.969	0.27	−4.78	1.51	0.986	0.29	−5.04
Rolling	1.75	0.967	0.40	−5.65	2.00	0.985	0.41	−6.11

Note. Bear TE: annualized TE (%) on days when the benchmark return is negative. Ratio: bear TE over overall TE. Stability: standard deviation of annual TE (%). Max DD: mean maximum active drawdown (%).

QIS is the most robust estimator: it dominates in the static framework, on short windows and in constrained universes. LW is very close, and takes the advantage when the covariance is persistent and on long windows. EWMA, designed for small universes and highly reactive covariances, is not suited to index replication on large universes. From an operational point of view, daily data are preferable to weekly data on liquid single-market equity universes, the exclusion rate is the main driver of tracking error, and quarterly rebalancing offers the best trade-off with turnover.

Limitations and extensions. The Factor Shock DGP relies on a single, strongly persistent calibration; a grid on ρ_B, σ_B and heavier tails ($\nu = 3$) would identify where LW and QIS exchange positions. The economic evaluation covers two liquid large-cap equity indices: adding a small universe (50 assets or fewer) and other asset classes (bonds, funds, multi-geography portfolios), where non-synchronous prices may favor weekly data, would test the transferability of the conclusions. Finally, a factor model (Fama–French, APT or PCA) would provide a useful structural benchmark against which to measure the marginal gain of nonparametric shrinkage.

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A. Parameters

Table 9. Parameters of the statistical evaluation

Parameter	Static Oracle	Factor Shock
Scenarios per configuration	10,000	1,000
Dimension	$p/T \in \{0.2, 1.0\}$	$N_{\text{sim}} \in \{100, 500\}$ assets
Sample length	$T = p/(p/T)$	≈ 805 days
Burn-in	60 observations	60 observations (excluded from losses)
Number of factors	10	10
True covariance	one per seed	AR(1) dynamics
Dynamics	—	$\rho_B = 0.95$, $\sigma_B = 0.05$, $\rho_d = 0.97$, $\sigma_d = 0.10$
Target daily volatility	1%	1%
Innovations	Gaussian, Student t_5	Gaussian, Student t_5
Windows	full sample	252, 504 days
Base seed	42	42

Table 10. Parameters of the economic evaluation

Parameter	Value
Indices	MSCI EMU ($\approx$ 240 stocks), MSCI USA ($\approx$ 600 stocks)
Data / evaluation period	02/01/2017–31/12/2025 / 2020–2025
Benchmark weights	quarterly capitalization weights, drifted daily
Rebalancing (historical backtest)	quarterly (index rebalancing date +1 day)
Windows	252, 504, 756 days (backtest); 126 to 756 days (validation)
EWMA decay	$\lambda = 0.94$
Constraints	full investment, long-only, exclusions
Solver	Clarabel (conic QP)
Transaction costs	none